

Stony Brook University
Instructor: Alejandro Melo Ponce

Fall 2016

MIDTERM EXAM II

NAME:

SBU ID #:

Instructions: Answer all three questions. The total number of points for the exam is 100. Each question has indicated the number of points it is worth. You should use the points as a guide for how to allocate your time during the exam. Answer each question only in the space provided. This exam is **open books/notes**. Calculators are allowed but other electronic devices are **not** allowed. For questions with a numerical answer, give a numerical answer.

You are **not** allowed to leave the room until you submit your exam.

GOOD LUCK!

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Problem 1 (40 pts)

There are two goods, food and clothing, whose quantities are denoted by x and y and prices p_x and p_y respectively. There is a consumer whose income is denoted by I and utility by U . His expenditure function is known to be

- (a) Find this consumer compensated and uncompensated demand functions.
- (b) Initially $I = 100$, $p_x = 1$ and $p_y = 1$. What quantities does the consumer buy, and what is his resulting utility?
- (c) Suppose that there is an increase in the price of food caused by the government levying a tax of 21% on it. How much will this consumer buy now and what is his resulting utility? What is the government revenue from this tax?
- (d) What is the welfare loss from this tax? Answer this question by calculating the compensating variation.
- (e) Consider the government revenue you calculated in part (c) and denote it by R^* . Suppose now that instead of levying a tax of 21% on food, the government decides to tax this individual via an income tax in the amount of R^* . This means that the consumer will have an after tax income of $I - R^* = 100 - R^*$, while facing prices of $p_x = p_y = 1$ and the government will still collect R^* in revenues. What are the consumer's optimal choices and his resulting utility?
- (f) Based on your answers to (e), is income taxation better than ad valorem taxation? Give the economic intuition behind this result. Ideally, you should argue your case by appealing to the Slutsky equation and the income and substitution effects.

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Problem 2 (40 pts)

Suppose that the utility function for goods x and y is given by:

$$U(x, y) = (x + 1)y.$$

1. Given a fixed level of utility, $\bar{U}$, graph the indifference curve. Based on your graph, explain why you might get a corner solution.
2. Calculate the uncompensated (Marshallian) demand functions for x and y . Be careful of the possibility for having a corner solution for x , i.e. $x^* = 0$ and explain what conditions need to hold so that you get a positive solution for x^* .
3. Describe how the Marshallian demand curve for x and y are shifted by changes in I or in the price of the other good.
4. From this point on, assume that $I \geq p_x$. Calculate the expenditure function for x and y .
5. Use the expenditure function in the previous part to compute the compensated demand functions for goods x and y . Describe how the compensated demand curves for x and y are shifted by changes in income or by changes in the price of the other good.

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Problem 3 (20 pts)

The compensated and the uncompensated demand, i.e. the Marshallian and Hicksian demand function, can be related through the Slutsky equation. The Slutsky equation can be derived from the equality

$$x(p_x, p_y, E(p_x, p_y, U)) = x^c(p_x, p_y, U). \quad (1)$$

1. State the Slutsky equation and briefly explain all the terms appearing in it.
2. Formally derive the Slutsky equation using equation (1).
3. Graphically illustrate the content that the Slutsky equation holds, carefully labeling all the objects that enter the graphic.

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